

11_pms_project_mgmt Project Management

Project Management

1 Formal Description

Identifier	Name	Owner	User ID
11_PMS_PM	Project Management	Hiroshi Amemiya Hiroshi.Amemiya@alpsalpine.com	la370177

Process Designers

- **Leader:** Hiroshi Amemiya Hiroshi.Amemiya@alpsalpine.com User ID: la370177
 - Reinmar Mueck reinmar.mueck@alpsalpine.com User ID: reinmar.mueck
 - Hidehiko Mitomo Hidehiko.Mitomo@alpsalpine.com User ID: I9970428
 - Akihide Yajima Akihide.Yajima@alpsalpine.com User ID: I9770666
 - Yoshihisa Takeda Yoshihisa.Takeda@alpsalpine.com User ID: JP003604
 - Hiromi Kato Hiromi.Kato@alpsalpine.com User ID: B0952028

1.1 Scope for Application of this process

In general and if not otherwise decided by Upper Management this process is applicable if one or more of the criteria below apply:

- any **OEM customer** projects
- all IVI development projects
- required A-SPICE
- ...

Applicability of this process defines that all activities are organized as a project. In this case this process also applies to all sub-projects. The decision to manage activities as a sub-project is recorded in the SoW.

Criteria to consider for the decision to break-down a project into sub-projects:

- are activities carried out by a different **project domain**
- **total effort** of the sub-project
- **number of resources** to be managed by the sub-project
- ...

1.2 Tailoring Guideline

The **project management** tailoring shall address differences with regards to

- project phases
- project levels (Project; Domain-Sub-project; Domain-Sub-sub-project)
- project domains

1.3 Relations

- Relations
 - Policies
 - Standards
 - A-SPICE 3.1
 - Fully covering: [MAN.3 Project Management](#)
 - Fully covering: [MAN.5 Risk Management](#)
 - Partly covering: [SUP.9 Problem Resolution Management](#)
 - Partly covering: [ACQ.4 Supplier Monitoring](#)
 - IATF 16949
 - To be specified in detail
 - ISO 26262
 - If functional Safety requirement(ASIL) are included in project scope, execute process which is defined in [FSQSX](#)
 - ISO/SAE21434
 - If Cyber Security requirement is included in project scope, execute process which is defined in [CSQSX](#).
- OTHERS
 - PMBOK (Edition 6)
- Unified QS
 - [MAN.3 Project Management](#)
 - [SUP.9 Problem Resolution Management](#)
 - [ACQ.4 Supplier Monitoring](#)

2 Strategy

2.1. Purpose

The purpose of the Project Management Process is to identify, establish, and control the activities and resources necessary for a project to produce a product, in the context of the project's requirements and constraints in a way that builds long-term stakeholder trust.

プロジェクト管理プロセスの目的は、長期的な利害関係者の信頼を造る方法で、プロジェクトの要件および制約に合わせて、プロダクトを作り出すために、プロジェクトに必要な活動および資源を識別、確立し、そして制御することである。

2.2. Objectives

- **Simplicity:**
 - Be easy to understand for stakeholders external to the organization (customer, suppliers)
 - Be easy to understand for new resources (hiring, out-sourcing)
 - allow globally staffed project teams
- シンプル:
 - 組織の外部の利害関係者（顧客、サプライヤー）にとって理解しやすいこと。
 - 新しいリソース（採用、アウトソース）にも理解しやすいこと。
 - グローバルなプロジェクトチームを可能にすること。

- allow project leaders with different project domain background to run the overall project
- allow switching the project leader role among different domains during the overall project life-time
- **Process change:**
 - allow coexistence of sub-projects which are following different project management approaches (e.g. Conventional vs. Agile)
 - allow for continuous process & tool improvements to be applied to running projects
- **Mastering complexity & change:**
 - manage projects of different level of size and complexity
 - manage projects with different content (from different project domains)
 - allow for changes in overall project structure
 - allow changes to project teams (supplier change)
- **Transparency:**
 - adjustable transparency for customer, suppliers and cooperation partners
- **Traceability**
 - traceability from requirements down to the single activity
- **Efficiency**
 - minimize human effort in any sense including calculation, traceability, reporting and monitoring
- 異なるプロジェクトドメインのバックグラウンドを持つプロジェクトリーダーが、プロジェクト全体を実行することを可能にすること。
- プロジェクト全体のライフサイクルの中で、異なるドメイン間でプロジェクトリーダーの役割を切り替えることができること
- **プロセスの変更:**
 - 異なるプロジェクト管理アプローチ（例えば、古典的とアジャイル）に従っているサブプロジェクトとの共存を可能にすること。
 - 継続的なプロセスとツールの改善を可能にし、プロジェクトの実行に適用することができること。
- **複雑さ、変化のマスター:**
 - 大きさや複雑さの異なるレベルのプロジェクトを管理すること。
 - 異なる内容のプロジェクトを管理すること。（異なるプロジェクトドメイン）
 - プロジェクト全体の構造を変更できるようにすること。
 - プロジェクトチームの変更を可能にすること。（サプライヤーの変更）
- **透明性:**
 - 顧客、サプライヤー、協力パートナーのための調節可能な透明性があること。
- **トレーサビリティ:**
 - 要件から単一の活動までのトレーサビリティを確保すること。
- **効率性:**
 - 見積もり、トレーサビリティ、報告、モニタリングなどあらゆる意味で人手による作業を最小限に抑えること。

2.3. Strategy

- utilize concepts and terminology established by PMBOK as one of the well known industry standards for describing project management
- define several levels for time-boxing (Project phase; Release Cycles; Domain phases; ...)
- unified milestone based planing (on overall project and sub-projects level)
- unified concept for managing scope of the project on top-level by deliverables (product and project deliverables)
- プロジェクト管理を記述するためのよく知られた業界標準の一つであるPMBOKでの確立された概念と用語を利用する
- タイムボックス化のためのいくつかのレベルを定義する（プロジェクトフェーズ、リリースサイクル、ドメインフェーズ、...）
- 統一されたマイルストーンベースの計画（プロジェクト全体とサブプロジェクトレベルでの）
- プロジェクトのスコープを成果物（製品とプロジェクトの成果物）別にトップレベルで管理するための統一されたコンセプト

- unified concept for abstracting various different task types by activities
- changes in scope, organization and responsibility are assumed the normal case not the exception
- continuously addressing stakeholder information needs at all levels of detail (flexible time consolidation of information; “poll and push” of information)
- アクティビティ別に様々なタスクタイプを抽象化するための統一された概念
- 範囲、組織、責任の変更は、例外ではなく通常のケースを想定する
- すべての詳細レベルで利害関係者の情報ニーズに対する継続的な対応（情報の柔軟な時間的統合、情報の「プッシュポーリング」）

3 Stakeholders

3.1 Skills, Capabilities, Responsibility & Authority

Note: Capability describes a maybe so far unrealized potential and is a combination of experience and intelligence. A skill you gain by a training.

Role		Skills	Capabilities	Responsibility	Authority
Project Manager	Project Leader	1.)PMP 2.)Common language with customer	1.)Negotiation and coordination with the customer 2.)Cordination with domain leaders 3.)Cordination with internal stakeholders	all deliverables of project	1.)authority to build project organization 2.)authority to assign domain leader 3.)authority to decide schedule and cost 4.)authority to manage a project
	Domain Project Leader	1.)Education program for domain 2.)Common language with customer of domain	1.)Expert knowledge of own domain 2.)Coordination with domain members	all deliverables of own domain	1.)authority to assign domain member 2.)authority to manage a domain 3.)authority to decide domain schedule and cost
	Module Leader	1.)Education program for domain/module 2.)Common language with customer of module	1.)Expert knowledge of own module 2.)Coordination with module members	all deliverables of own module	1.)authority to assign module member 2.)authority to manage a module 3.)authority to decide module schedule and cost

3.2 Stakeholder involvement

Meeting	Method /Type	Frequency	Role 1	Role 2	Role 3	Role 4	Project Leader
WS X	WS at supplier	Twice			R		
Review Y	Webex	Weekly	A	AR	I	I	I
CCB Meeting	Phone Call	Weekly starting 2016-09-03				R	
Regular Management Review	Webex	Weekly					
Kick-off	FTF MTG						
MLA Q-Gate	FTF MTG						
Quality Gate	FTF MTG						
<u>Responsible, Approving Authority, Support, Informed and Consulted)</u>							

4 Input Work products

- [Project Statement of Work \(SoW\)](#)
- [Master Schedule](#)
- [Organization Chart](#)
- [Deliverable Rollout Plan](#)
- Gate Review Record
- [Lessons Learned Record](#)
- Resource Calendar
- [Cost Estimation Sheet](#)
- Requirement Analysis Result
- [Communication Management Plan](#)
- Task Ticket
- Training Ticket
- Risk Ticket
- Progress Report

5 Output Work products

- [Project Statement of Work \(SoW\)](#)
- [Schedule Management Plan](#)
- [Master Schedule](#)
- [Organization Chart](#)
- [Scope Management Plan](#)
- [Deliverable Rollout Plan](#)
- [Acquisition Management Plan](#)
- [Resource Management Plan](#)
- Resource Calendar
- [Cost Management Plan](#)
- [Cost Estimation Sheet](#)
- [Cost Baseline](#)
- Budget registration sheet
- [Communication Management Plan](#)
- Lop Ticket
- Task Ticket
- Training Ticket / Training Record
- [Risk Management Plan](#)
- Risk Ticket
- Equipment purchase order sheet
- Progress Report
- [Lessons Learned Record](#)
- Gate Review Record
- Evaluation Chart

6 Process Practices

6.1 Life Cycle relation

The project management process shall support several project life cycles as well as their combination.

- **Conventional:** for complicated problems that can be solved by the company's experience and that are predictable and plannable in terms of most unknowns that could influence the timely completion of defined scope. Depending on project duration and size the planning can be fine granular from the beginning or detailed by rolling wave planning during project execution. Following the development process of our customers that foresee a stepwise extension and improvement of the deployed and validated work product the iterative and incremental lifecycle by e.g. I-Step is most common.
- **Adaptive (using Agile_flow methods):** for complex problems that are un-predictable for any reason and don't allow a proper planning of unknowns or predefined activities that should lead to its solution. For this purpose the software development department implemented the Scrum framework and its scaled extension framework LeSS. It is fully compatible to Safe5.0 framework on "Team Level" and can be extended/tailored to fit into customer's "Program Level" as well.

Typically the overall multi-phase project is organized following an iterative and incremental life cycle. Independently from selected lifecycle model the iterative phases follow the principle of the Denim cycle (see 6.4 Process Flow) and it's steps of **P**lanning, **D**oing, **C**hecking and **A**cting. Between each phase transition closing and initiation procedures are obligatory. The iteration cycle defined in the Scrum framework is called Sprint and is following this principle to create a work product (Sprint Goal), as well, but typically with a higher frequency as an sub-ordinate activity within a higher project level iteration (later called Release Cycle).

6.2 Sequence & interaction with other processes

As the project management processes provides the framework for managing ultimately all activities within a project including it's sub-projects, this process has by far the most interrelation with other processes. Therefore many other process activities are triggered by "request signals" from project management.

The activities as described by other processes are planned and managed as described within this process in general. For planning and managing any activities there are always two options:

- **Option 1:** Organized as **seperate Sub-(Sub)-Project**. This means at least the Project Management process has to be carried out separately, but in coordination with the parent project
- **Option 2:** Organized as **part of the overall Project or any Sub-Project**. *Note:* Reporting and Escalation path can be different e.g. to allow for independence of SQA for example.

Also Agile Sub-Projects have to sync with the normally plan-driven parent project or other sub-projects in order to resolve dependencies and to allow planning on a business goal, feature or epic level. As a detailed planning below epic level will only be performed towards the next sync point (planning Milestone) dependencies to agile teams should be reduced to an absolute minimum. Therefore it makes sense to form agile and non-agile teams as (vertical) feature teams instead of (horizontal) component teams.

6.3 Entry & Exit conditions (triggers)

• Start triggers

- project phase transfer
- Complete plan and preparation for each domain
- Completion of the release cycle for each domain

• end triggers

- Trigger to elicit raw requirement
- Define KPI
- Request domain specific Plan
- create supplier management
- 21_elicit_stakeholder_requirement started
- Identify Risk
- Change definition KPI
- Measure KPI
- Start Defect management process
- Start Issu management process
- Start change request management process
- Completion of the Development Phase
- project phase transfer

• 開始トリガー

- プロジェクトフェーズへの移行
- 各ドメインの計画および準備の完了
- 各ドメインのリリースサイクルの完了

• 終了トリガー

- 利害関係者要求の受け取り
- KPIの定義
- ドメイン計画の要求
- サプライヤー管理計画の作成
- 21_elicit_stakeholder_requirement開始
- リスクの特定
- KPIの変更
- KPIの測定
- Defect管理プロセスの実施
- Issu管理プロセスの実施
- 変更依頼管理プロセスの実施
- Development Phase の完了
- プロジェクトフェーズへの移行

• Start Conditions

- The “Project Request”-Phase, which is optional, is the first phase the project management process is formally started as per management decision. During this phase the project management process typically applies only partly (refer to the respective tailoring)
- With the “Project Request Phase”, typically the project starts and therefore the project management process is started.
- Note: Different project types will require specific conditions. (refer to the respective tailoring)

• Exit Conditions

- **Normal Case:** The project management process activities do end with the successful execution of the last release cycle. In particular the required management approval as part of the “Closing” activity have to be received.
- **Exceptional Case:** A release cycle or in extreme a project can be stopped always by management decision, which is basically pulling up the management decision during the “Closing” activity of each of the release cycles. If the decision is to finalize the project executing the “Closing” activity is mandatory. (Note: A Release Cycle is sometimes not finalized in the “Project Preparation Phase” if during an RfQ the circumstances do change drastically.)

Domain Specific Phases

SW Domain Phases

Considering the development process in the automotive industry, that requires a reliable and matured product in order to warrant a safe transport, the development phases in the **SW Domain Project** are partitioned into the following **SW-Domain Phases**

- **Architecture Phase**
- **Implementation Phase** (AKA: Construction Phase)
- **Maturity Phase** (AKA: Bug-fixing Phase)
- **Stability-Phase** (AKA: Show-stopper Phase, which is too negative)

Each SW-domain phase consist of one or several release cycles, which have to be defined by each project.

On the first glance this seems to contradict the idea of Scrum to develop a minimum viable product (MVP) with reduced time to market that is extended step by step with additional features and improved based on market feedback, but if the MVP is considered the minimum agreed scope for SOP and the bug-fixing phase a kind of market feedback from the system testers/engineers the framework still can be applied. The initial product architecture is based on the knowns in the architecture phase and will be adapted during refinement of the user stories in the agile process during implementation phase. The architecture has to ensure that the fixed requirements are not violated. Low level technical requirements can be added during the implementation phase. Change Requests apply from Epic level and above.

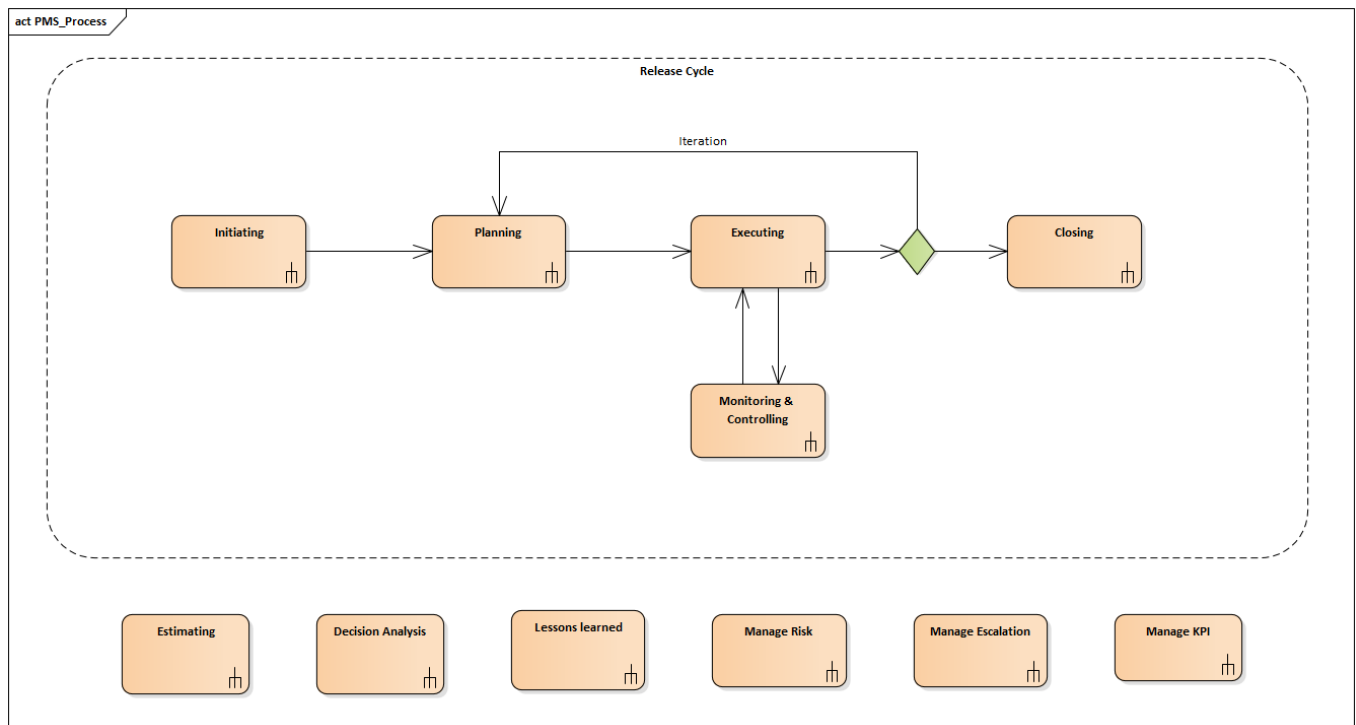
Nevertheless the automotive release procedure requires a phase of stability before market introduction that is used to obtain legal approval and certifications from Licensees (non-regulatory) as well as governments (regulatory).

After this gate is passed additional features and updates can be released mainly on application level into the market that do not affect the product's accreditation.

6.4 Process Flow

The PMS process consists of the following Main activities, which are executed in each Release Cycle in a sequential order:

- **Initiating** - to establish and maintain the overall framework for planning the project or phase
- **Planning** - to identify the deliverables, to establish the schedules and to plan and assign each activity
- **Executing** - to carry out the activities as planned
- **Monitoring & Controlling** - to check the actual performance against the established plans and to react accordingly
- **Closing** - to review the phase or project high- and low-lights in order define activities to continuously improve as well as to perform necessary closing activities



In addition there are activities which are in parallel executed and triggered from the above

- **Estimating**
- **Decision Analysis**
- **Lessons Learned**
- **Manage Risk**
- **Manage Escalation**
- **Manage KPI**

7 Infrastructure

tool,Mechanisms	purpose	TargetActivity
Redmine.connect	Project Management	Scope,Schedule,Cost,Risk,List of open items,Issue,Problem,Task,Userstory,etc
Dokuwiki	Process Description	Reference of Standard Process and Process definition of MARS
Scheduling Tool	Project Planning	Scheduling e.g)MS-Project
Mailing Tool	Communication	Communication e.g)Ms-Outlook

8 Performance Measurements

🔧 **To describe here:** Which KPIs are defined for measuring the process performance

- Monitoring
 - Effectiveness KPI
 - Efficiency KPI
- Qualitative & quantitative measurements

🔧 **Fix Me!** Describe all the usual KPIs implemented by the Standard Tool ... 😊

- Quality of Estimates (effort; duration; cost)
- Work load of resources (overload)

9 Review

🚨 **To describe here:** Which reviews exist for checking the process performance (e.g. peer reviews) and which Review Criteria are defined for each review.

- Review Name 1
 - Review Criteria
- Review Name 2
 - Review Criteria

10 Quality Gates

Responsibility

The responsibility for performing **quality gates** as defined is with the overall project leader. If the **Project Management** process is carried out on sub-project level the sub-project leaders responsibility is to provide the required input and to participate in it's parent project's Quality gate.

Quality Gate input list for Project Management

- All deliverables from Project Management with evidence of acceptance as planned in the Delivery Rollout Plan for the current phase.

11 Functional Safety

If functional Safety requirement(ASIL) are included in project scope, execute process which is defined in **FSQSX**

Project Management Process of FSQSX are defined in **FSQSX-MAN (en)** / **FSQSX-MAN (jp)**.

By Executing FSQSX activities, following SPW WorkProduct and activities will be influenced.

- WorkProduct
 - [Project Statement of Work \(SoW\)](#)
 - [Deliverable Rollout Plan](#)
 - [Additional Workproducts - Functional Safety Relevant](#)
- Standard Role
 - [Functional Safety](#)

12 Cyber Security

If Cyber Security requirement are included in project scope, execute process which is defined in **CSQSX**

Project Management Process of CSQSX are defined in **CSQSX-MAN.en** / **CSQSX-MAN.jp**.

By Executing CSQSX activities, following SPF WorkProduct and activities will be influenced.

- WorkProduct
 - [Project Statement of Work \(SoW\)](#)
 - [Deliverable Rollout Plan](#)
 - [Additional Workproducts - Cyber Security Relevant](#)
- Standard Role
 - [Cyber Security](#)

13 TISAX

TISAX(Trusted Information Security Assessment Exchange) is an assessment and exchange mechanism for information security in the automotive industry. In recent years , OEMs require suppliers to obtain TISAX certification.

ALAP defines [ISMS standards](#), and this ensures that TISAX requirements are to be complied with to some extent, but additional TISAX-specific requirements must be addressed. The project should analyze the TISAX requirements and take action to comply with the requirements.

- Examples of project activities requiring TISAX :
 - Vulnerability check of tools used for external access.
 - Conduct assessments for suppliers with external access.(It must be done by the tools used)
 - Conclude an NDA with individual members participating in the project.

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